

Cloud Computing Course

Assignment 2 – GCP Services & Cloud Fundamentals

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Submission Instructions

1. Create a single PDF document for submission.
2. Answer theory questions using the corresponding question numbers.
3. For practical questions, attach screenshots showing configuration and successful output.
4. Submit a single consolidated PDF.
5. Total Marks: 50 + 10 Bonus Marks.

Section A – Theory (10 Marks)

- Q1. Define Cloud Computing in your own words. (2 Marks)
- Q2. Differentiate between IaaS, PaaS and SaaS with one example each. (3 Marks)
- Q3. Explain Region and Zone. Why are multiple zones important? (2 Marks)
- Q4. Explain IAM and the difference between Viewer, Editor and Owner roles. (3 Marks)

Section B – Compute Engine & Networking (15 Marks)

- Q5. Create a Compute Engine VM and attach screenshots of VM creation, running VM and SSH terminal. (3 Marks)
- Q6. Record VM Name, Region, Zone, Machine Type, Internal IP and External IP. Explain each in one sentence. (2 Marks)
- Q7. Create a Network Tag named mysql-server and create a firewall rule allowing TCP Port 3306. Attach screenshots. (4 Marks)
- Q8. Create a Network Tag named postgres-server and create a firewall rule allowing TCP Port 5432. Attach screenshots. (4 Marks)
- Q9. Explain the purpose of firewall rules and network tags in GCP. (2 Marks)

Section C – Cloud Storage (10 Marks)

- Q10. Create a Cloud Storage Bucket and attach screenshots. (2 Marks)
- Q11. Upload at least three files of different types and attach screenshots. (3 Marks)
- Q12. Explain the difference between Object Storage and Traditional File Systems. (2 Marks)
- Q13. Record the Bucket Name, Region and Storage Class. Explain each in one sentence. (3 Marks)

Section D – Pub/Sub (5 Marks)

- Q14. Create a Topic and Subscription. Attach screenshots. (2 Marks)
- Q15. Publish a message and retrieve it using the Subscription. Attach screenshots showing successful delivery. (3 Marks)

Section E – BigQuery (10 Marks)

Q16. Create a Dataset in BigQuery and attach screenshots. (2 Marks)

Q17. Upload a CSV file into a table and attach screenshots. (2 Marks)

Q18. Run: `SELECT * FROM table_name LIMIT 5;` and `SELECT COUNT(*) FROM table_name;`
Attach screenshots. (4 Marks)

Q19. Explain the difference between a Dataset and a Table in BigQuery. (2 Marks)

Bonus Challenge (10 Marks)

Design a cloud architecture for an Online Food Delivery Application using Compute Engine, Cloud Storage, Pub/Sub and BigQuery.

Show the complete flow from customer order to analytics dashboard.

Label all services and explain the role of each service.